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Key Points:
- The Permian paleo-position of Indochina and its drifting history in the Paleo-Tethys Ocean is constrained paleomagnetically
- The model of Permian Cathaysialand in the Paleo-Tethys Ocean is supported quantitatively
- The average rate of Indochina drifting away from Gondwana is estimated to be ~4.5 cm/year during Early-Middle Permian to Late Triassic

Supporting Information:
- Supporting Information S1
- Table S1

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Paleomagnetic Study on the Permian Rocks of the Indochina Block and Its Implications for Paleogeographic Configuration and Northward Drifting of Cathaysialand in the Paleo-Tethys

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Abstract The drift of Gondwana-derived blocks across the Tethys has remained unclear specifically because one of the largest components, Indochina, has remained unconstrained by paleomagnetism during the late Paleozoic, the interval comprising the first half of its journey towards the Eurasian landmass. Hence, we report a paleomagnetic study of Early-Middle Permian limestones from central Thailand yielding a high-quality data set for the Indochina Block with probable primary origin. The magnetization yields a paleomagnetic pole at 34.1°N, 331.7°E ($K = 45.4$, $A_{95} = 5.7°$) corresponding to a paleolatitude of 21.2 ± 5.7°S at the center of the study area (15°N, 101°E) and supporting a tropical restoration for late Paleozoic Indochina. A total of ~5,000 km northward transition from a proximity to Gondwana in the Early-Middle Permian (ca. 280 Ma) to the southern margin of Eurasia by the Late Triassic (Norian, ca. 220 Ma) is identified and yields an average northward latitudinal drift rate of ~4.5 cm/year. A concurrent northward drifting history during the Permian is suggested for the Indochina, North Qiangtang, and South China blocks, and thus, the widely used model of Permian Cathaysialand is quantitively supported when combined with their paleobiogeographic affinities.

1. Introduction

The Tethyan drift history of the Gondwana-derived blocks towards the Eurasian landmass has been a hotly debated topic for decades (Cocks & Torsvik, 2013; Collins, 2003; Ferrari et al., 2008; Metcalfe, 1986, 1996, 1999, 2001, 2002, 2011a, 2011b, 2013; Scotese, 2004; Torsvik & Cocks, 2013). The Indochina (or Annamia) Block is considered to be one of the major blocks to separate from eastern Gondwana and drift northward and amalgamate with the other eastern Asia blocks (e.g., North China and Tarim blocks) along the southern margin of Eurasia during the Mesozoic era. The late Paleozoic paleo-position and drift history of the Indochina Block has remained one of the biggest unresolved problems in the paleogeographic reconstructions of eastern Asia within the Gondwana and Pangea supercontinent assemblies (Cocks & Torsvik, 2013; Collins, 2003; Metcalfe, 2011a, 2011b, 2013; Scotese, 2004; Torsvik & Cocks, 2013). In some reported reconstructions, Indochina is considered to be similar paleobiogeographically with the South China Block, implying a continental connection between them; a super-terrane is thus believed to exist in the Paleo-Tethys during the Carboniferous-Permian and termed Cathaysialand (Metcalfe, 2009; Wu et al., 1995). Although little is known on the Paleozoic position of the Qiangtang blocks, studies on the late Paleozoic sedimentary sequences and fossils also suggests a Cathaysia-affinity for the North Qiangtang Block (Jin, 2002; Zhang et al., 2009). The widely referred but yet to be confirmed Cathaysianland model is primarily attributed to tectonostratigraphic and paleontological evidences from the Paleozoic to Mesozoic rocks (Metcalfe, 2001, 2011a, 2013), although conflicting interpretations are also acknowledged (Fang & Cope, 2004).

Previous paleomagnetic work on Paleozoic rocks of the Indochina Block has been carried out in the Loei-Nam Nao area of eastern Thailand by Chen and Courtillot (1989) and Yang and Besse (1993), in northwestern Thailand by Bhongsuwan and Elming (2000), and in southern Peninsular Thailand by Bhongsuwan and Elming (2000) and Richter and Fuller (1996). All of these studies resolved magnetizations that were either of syn-folding or otherwise of overprinted origin. The implication is that the first half of the journey of the Indochina Block from Gondwana to Eurasia is still unconstrained by paleomagnetic control. This in turn

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leads to a discrepancy of the position of Indochina in the two contrasting reconstruction models, namely, an equatorial configuration that locates Indochina at the Equator (Collins, 2003; Ferrari et al., 2008; Metcalfe, 1996, 1999, 2001, 2002, 2011a, 2011b, 2013) or a tropical configuration locating Indochina at middle-low southerly latitude (Cocks & Torsvik, 2013; Scotese, 2004; Torsvik & Cocks, 2013; Ueno, 2003).

In this study, we reported a new paleomagnetic study from Permian limestones in central Thailand and combine the results with reliable previously reported Mesozoic results from Indochina with the aim of resolving the drift history of Indochina across the Paleo-Tethys.

2. Geologic Settings and Sampling

The Indochina Block is one of the largest blocks in eastern Asia and occupies most of the Southeast Asia; it has formerly been named the Indosinian Craton (Bunopas & Vella, 1978) although the nature of much of the continental foundation remains unclear. Its boundary with the western Sibumasu Block was once regarded as the Nan-Uttaradit lineament (Barr & Macdonald, 1987; Bunopas, 1981), until the Sukhothai Terrane between the Nan-Uttaradit suture and Chianglai-Chiangmai suture was recognized as an arc-related terrane distinct from the remainder of Indochina (Barr & Macdonald, 1991). The Nan-Uttaradit Suture subsequently lost its status as the closed remnant of the main Paleo-Tethys Ocean and is now increasingly accepted as a back-arc basin. The boundaries between Indochina and the other blocks (e.g., Song Ma suture to South China, Bentong-Raub suture to East Malaya, Median Sumatra suture to West Sumatra) have become basically clear following the work of Ueno (2003), Sone and Metcalfe (2008), and Ueno et al. (2012), although a contrasting assessment has been given by Cai and Zhang (2009).

The study area, petrographically the Khao Khwang Platform, geotectonically is the western margin of the Indochina Block. It comprises the area to the east of the Nan-Uttaradit Suture Zone, north of Mae Ping Fault, and west of the Nam Duk Basin (Figure 1; Ridd et al., 2011). The Permian succession in the Saraburi area comprises the Saraburi Group and incorporates the Sap Bon, Khao Khad, Pang Asok, Nong Pong, Khao Khwang, and Phu Phe formations. In the Nakhon Sawan-Phetchahun area the Permian succession comprises the Tak Fa Formation (Figure 2). Note that these formations are nearly contemporaneous as shown in Figure S1 in the supporting information, rather than in a continuous succession (Ridd et al., 2011). A total of 280 cores in limestone facies were drilled from 27 sites through five sections using a portable gasoline-powered and water-cooled hand-drill. Field samples were oriented using a magnetic compass, or both magnetic and sun compasses when feasible. Both orientation methods were applied to 36% of the samples and identified local geomagnetic declinations of $-1°$ in the Nakhon Sawan-Phetchahun area and $-2°$ in the Saraburi area.

The Tak Fa Formation is mainly massive to well-bedded limestone with subordinate sandstone and shale, which stretches from north of Lopburi Province, through the type area at Tak Fa village in southeast Nakhon Sawan Province northeastwards to Phetchabun Province (Figure 2; Figure S2a; Ridd et al., 2011). Three sites (Th218, Th219, and Th220) were collected from section 1 (road cut) located southeast of Tak Fa village, and four sites (Th221, Th222, Th223, and Th224) and five sites (The225, The226, Th227, Th228, and Th229) were respectively collected from section 2 (natural section) and section 3 (quarry section; Figure S2b) at the northwest of Bang Sam Phan village. The age of these three sections of the Tak Fa Formation is paleontologically constrained to the late Early Permian to early Middle Permian (Chitnarin et al., 2012; Metcalfe & Sone, 2008), while the Khao Khwang and Nong Pong formations in the Saraburi area are Early-Middle Permian (Cisuralian-Guadalupian) in age (Fontaine, 2002; Ridd et al., 2011; Udchachon et al., 2014). The Khao Khwang Formation is dark to light gray, thick bedded limestone with nodular chert (Figure S2c and S2d), which occurs in the northern part of the Saraburi area, and the Nong Pong Formation is laminated to thin or thick bedded shale and thin bedded limestones with locally developed chert, distributed widely in the central part of the Saraburi area (Figure 2; Ridd et al., 2011). Sites Th392-402 and Th403-406 were collected from section 4 (road cut) of the Khao Khwang Formation and section 5 (road cut) of the Nong Pong Formation, respectively (Figure 2).

The Permian strata are separated from the Mesozoic sediments by the Indosinian I Unconformity, which was formed during the most intense episode of the Indosinian Orogeny and corresponded to the great Upper Permian to Middle Triassic hiatus. Most Paleozoic strata were deformed or folded during the Indosinian I Event, although its actual timing cannot be closely constrained (Ridd et al., 2011).

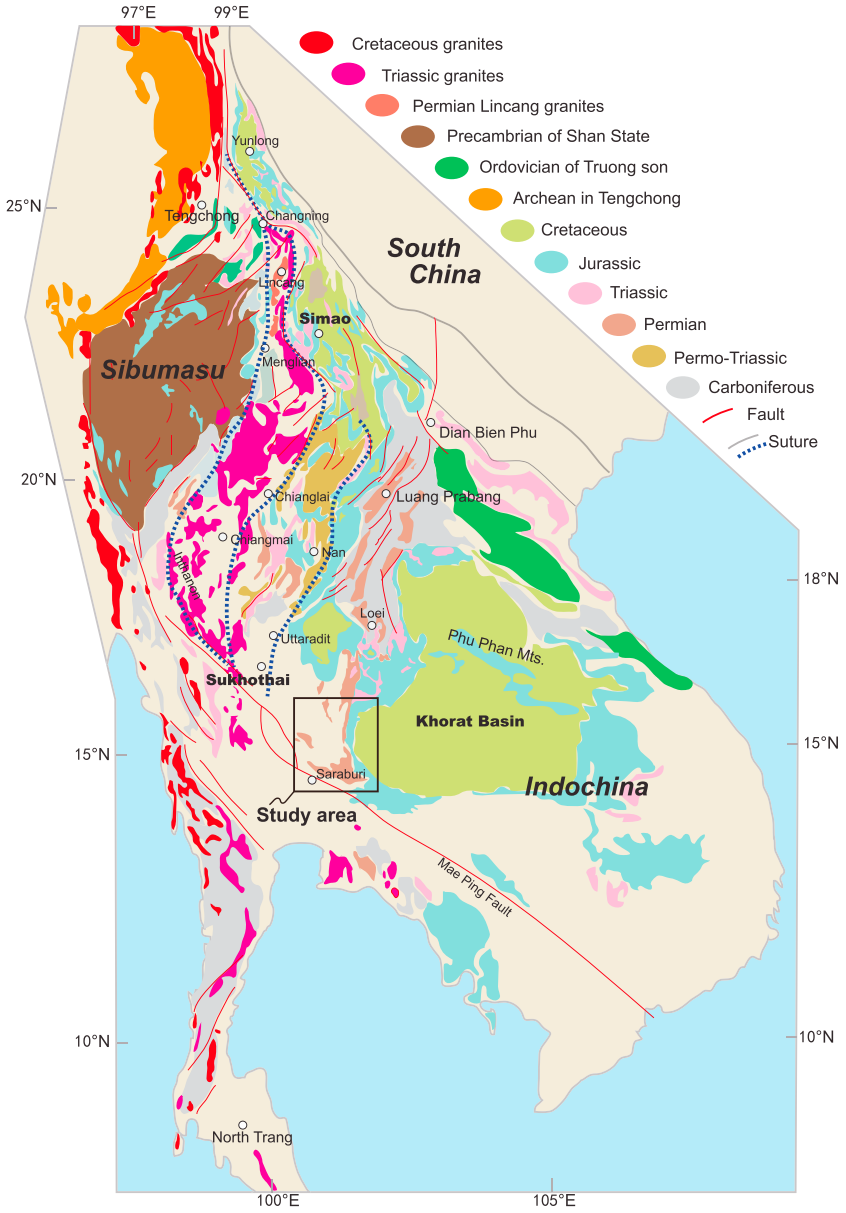


Figure 1. Schematic geologic map of East Asia and surrounding regions, revised from 1:1,500,000 geologic map by the Institute of Chengdu geology and minerals in 2006.

3. Paleomagnetic Analysis

3.1. Laboratory Methods

In the laboratory, the core samples, generally 2.5 cm in diameter and 6–13 cm in length, were trimmed into standard cylindrical specimens; ~2.2 cm in length with long samples were cut into two specimens designated S_i (inner) and S_o (outer). Systematic rock magnetic analyses were conducted on samples powdered from representative fresh end material to identify the dominant magnetic carriers. Acquisition and back-field demagnetization of IRM (isothermal remanent magnetization) and hysteresis loops were measured with a Micro 3900 vibrating sample magnetometer (3900VSM) at room temperature with an applied field range of ±1.0 or ±1.5 T. For samples not saturated at up to 1.5 T, acquisition of IRM at up to 2.7 T was measured on cylindrical specimen S_o (while specimen S_i cut from the same sample reserved for paleomagnetic measurement) using a 2G-755R cryogenic magnetometer and 2G Enterprises pulse magnetizer. Temperature dependence of the magnetic susceptibility was determined in argon from

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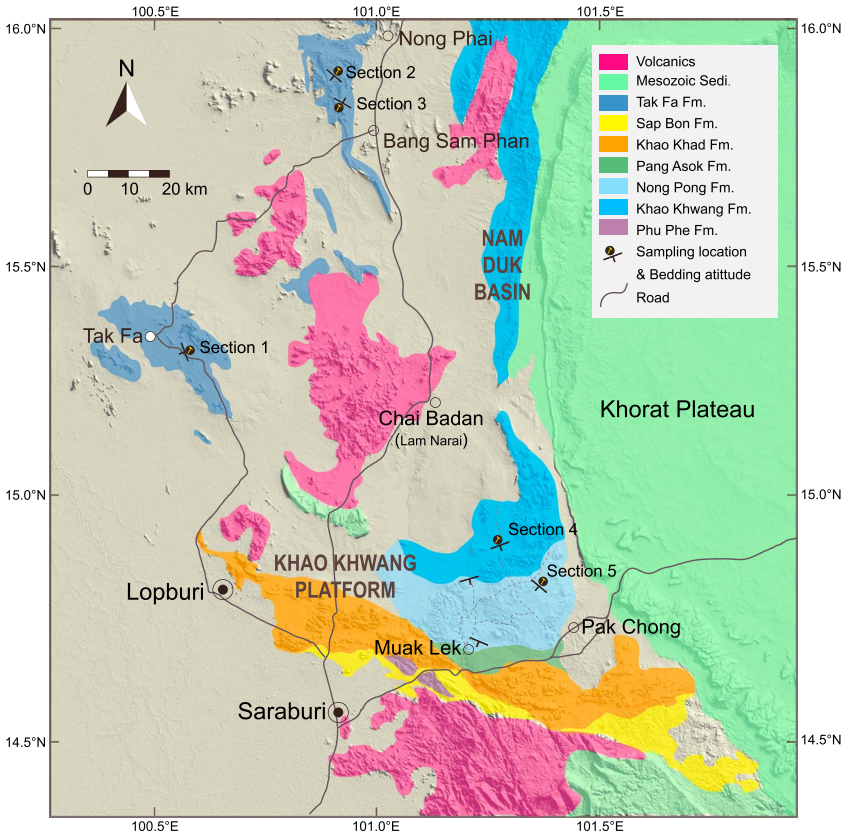


Figure 2. Geological map of Permian strata in the study area in central Thailand (revised from 1:5,000,000 geological map of Thailand, DMR, 2011; Ridd et al., 2011).

room temperature to 700°C using an AGICO multifunction Kappabridge (MFK1). Representative cylindrical S_o specimens were used for three-axis IRMs (Lowrie, 1990) and magnetized with 0.1, 0.5, and 2.5 T along the X, Y, and Z axes respectively using a 2G Enterprises pulse magnetizer and then thermally demagnetized using an TD-48 demagnetization oven and measured with a 2G-755R cryogenic magnetometer. In addition, low temperature measurement of IRM was made in the range of 15–300 K to identify the diagnostic Verwey transition for magnetite, the Morin transition for hematite, and the 34 K transition for pyrrhotite, using a Magnetic Property Measurement System-Extra Large 5 Superconducting Quantum Interference Device (MPMS-XL5SQUID). Samples were first cooled from 300 to 15 K in zero field (ZFC), followed by applying a 2.5 T field at 15 K, and measured from 15 to 300 K. The cooling-warming cycle of a saturation isothermal remnant magnetization (SIRM) imparted at 300 K with 2.5 T was measured in a zero-magnetic field.

Following rock magnetic analyses and the measurements of natural remanent magnetization, specimens were subjected to low temperature thermal cycles in liquid nitrogen for two or three times as suggested by Kirschvink et al. (2015) with the aim of erasing viscous components, which might be carried by multidomain magnetite. Thermal or AF (alternating field) demagnetization was then applied to samples using a 2G-RAPID cryogenic magnetometer in the Paleomagnetism and Geochronology Laboratory (PGL) of the Institute of Geology and Geophysics, Chinese Academy of Sciences. The magnetometer is installed in a magnetically shielded room with a background field weaker than 300 nT. Thermal demagnetization was conducted using a PGL-designed TD-PGL-100 oven with internal residual magnetic field lowered to less than 5 nT (generally 1–2 nT).

The methods of principle component analysis (PCA; Kirschvink, 1980) or great-circle fitting (GC; McFadden & McElhinny, 1988) were used in the calculation of demagnetization results, and Fisher statistics (Fisher, 1953) were used in the mean direction calculation.

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3.2. Rock Magnetic Results

Different magnetic behaviors of samples from the three formations can provide important information on the compositions of the dominant magnetic minerals.

For the Tak Fa Formation samples, the IRM saturated before 0.2 T (Figures 3e–3g) and the hysteresis loop closes before 0.2 T (Figures 3b and 3c) indicating a relatively low coercivity (or an admixture with paramagnetic components like sample Figure 3a) less than 50 mT and centralizes at ~20 mT. Combined with the record of the Verwey transition (Muxworthy & McClelland, 2000; Özdemir et al., 2002; Verwey, 1939) identified in the ZFC (Figure 3h), the dominant magnetic carrier in the Tak Fa Formation samples is interpreted to be magnetite.

The Khao Khwang Formation samples have a dominant soft component (less than 0.1 T) after the hard component (0.5–2.5 T) is erased before 100°C (Figure 3j), which decreases toward zero after 500°C (Figure 3k). The differential of IRMs indicate a very high coercivity component and a component with low coercivity less than 50 mT showing a tendency to saturation before 1.0 T (Figure 3i). Together with the unblocking temperature at ~580°C (Figure 3l), the dominant magnetic minerals in the Khao Khwang Formation samples appear to be a mixture of magnetite and goethite. This interpretation is also confirmed in the subsequent thermal demagnetizations, which reveals a sharp drop before 120°C and a stable component above this temperature (Figures 4e–4g).

For the Nong Pong Formation samples, the IRM saturates after 1.5 T, whose differential reveals two partially overlapping coercivity components centered at less than 100 and ~200 mT, respectively (Figure 3m). The thermal demagnetization of the three-axis IRM shows that the soft component (less than 0.1 T) decreases toward zero after 500°C, after the medium component (0.1–0.5 T) has dropped sharply between 300–350°C (Figure 3q). Associated with the Verwey transition and ~34 K transition (Dekkers et al., 1989; Rochette et al., 1990; Wolfers et al., 2011) identified in RT-SIRM (Figures 3n and 3p) and ZFC (Figures 3r and 3t), fine grain magnetite and pyrrhotite are identified to be the dominant magnetic carriers in the Nong Pong Formation samples.

3.3. Paleomagnetic Results

3.3.1. Tak Fa Formation

One hundred twenty-three field samples from 12 sites were collected from the Tak Fa Formation, and two groups of characteristic remanent magnetizations (ChRMs) denoted Groups 1 and 2 were isolated. In Group 1 only one component can be isolated during the AF demagnetization up to 110 mT (Figure 4b) with the directions of samples from sites Th220, Th221, Th222, Th223, and Th224 clustering in the location of the present geomagnetic field and dispersing following tilt adjustment (Figure S1 and Table S1). Hence, Group 1 is interpreted to be an overprint in the present geomagnetic field (Figure S1a). Group 2 comprises four sites, including one with reversed polarity (Th219) and three with normal polarity (Th226, Th228, and Th229; Figure 5a). Two contrasting component directions were isolated from these samples (Figures 4a, 4c, and 4d) with the high field component ($>$20 mT) interpreted as the ChRM.

3.3.2. Khao Khwang Formation

One hundred fourteen samples from 11 sites were collected from the Khao Khwang Formation. Following a sharp drop of the magnetization between natural remanent magnetization and 150°C probably corresponding to the presence of goethite, the low temperature component was erased before ~350°C, and the stable component was isolated between ~350 and 520°C or higher and interpreted as the ChRM. Note that the magnetization had dropped to several 10^{-6} A/m (corresponding to the magnetic moment of standard specimens in 10^{-8} emu) after removal of the low temperature component (Figures 4e, 4f, and 4g). Isolation of the very weak but stable high temperature component benefited from the optimized 2G-RAPID magnetometer and especially the good performance of the TD-PGL-100 oven. As a result, ChRMs could be isolated from seven sites in this formation in section 4 (normal polarity: Th392, Th394, Th397, Th398, Th401, and Th402; reversal polarity: Th395; Figure 5b).

3.3.3. Nong Pong Formation

Forty field samples from four sites were collected from the Nong Pong Formation and following removal of a low temperature (or field) component below 150°C (or 20 mT); a stable medium temperature component was isolated between 200 and 430°C or higher (Figures 4h and 4i). As noted in section 3.2, pyrrhotite is one of the main magnetic carriers; this has a Curie temperature no higher than 325°C (Dekkers et al., 1989)

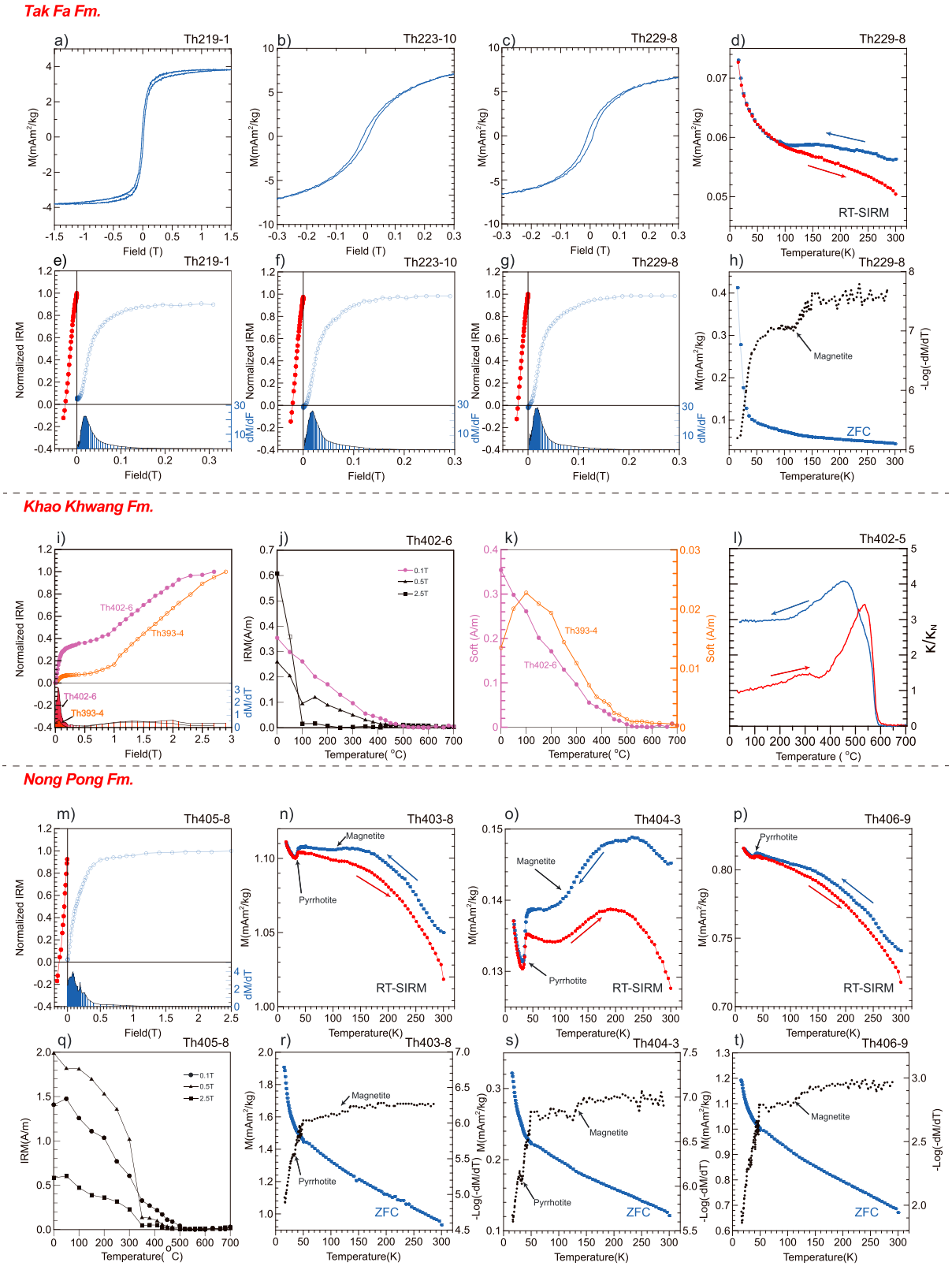


Figure 3. Rock magnetic analysis of the representative samples of the Tak Fa, Khao Khwang and Nong Pong formations (Fm.). (a, b, and c) Hysteresis loop; (e, f, g, i, and m) reversal field demagnetization, IRM (isothermal remanent magnetization) and its first-order derivative (blue histogram); (d, n, o, and p) RT-SIRM variation during low temperature cycles; (h, r, s, and t) ZFC variation during low temperature cycles; (j and q) thermal demagnetization of three-axis IRM; (k) the soft components during the thermal demagnetization of three-axis IRM; (l) normalized susceptibility versus temperature. See section 3.2 in text for more discussion.

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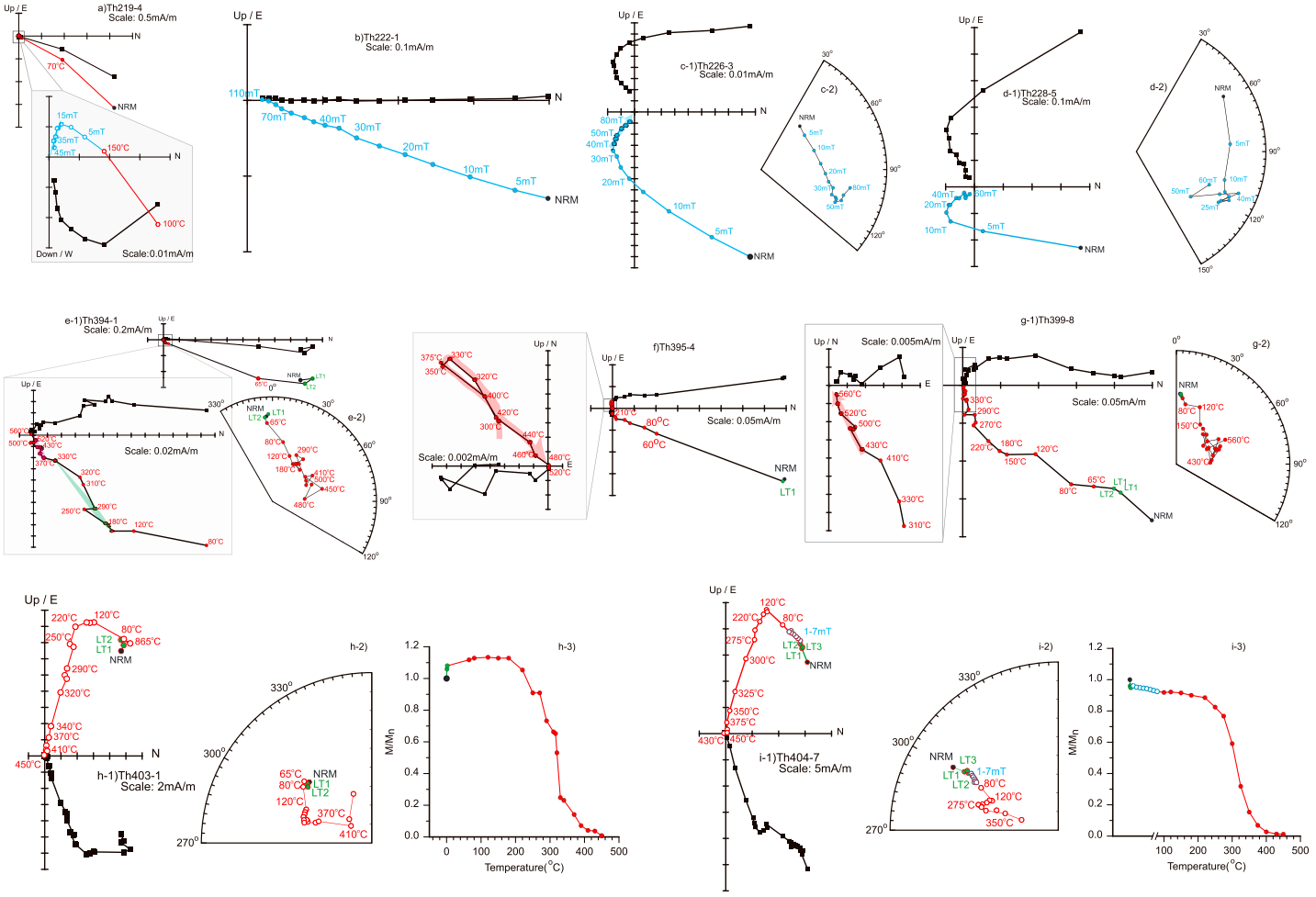


Figure 4. Demagnetization diagrams of representative specimens. AF (alternation field) demagnetization, marked with blue color; TD (thermal demagnetization), marked with red color; low temperature thermal cycles in liquid nitrogen, marked with green color; solid (open) circles corresponds normal (reversal) polarity in vertical projection; black squares represent horizontal projection. Magnetic directions are also plotted in equal-area projections. (h-3 and i-3) Normalized magnetization variation during heating, M = magnetization at every single step, Mn = natural remanent magnetization. All Zijderveld and equal-area projection diagrams are plotted in geographic coordinate.

and evidently contributes most to the medium temperature component. Note that the magnetization direction between 200 and 325°C (pyrrhotite bearing) is consistent with that between 325 and 430°C or higher (magnetite bearing; Figures 4h-1 and 4i-1); the pyrrhotite and magnetite apparently record the same magnetic signal, and further statistical analysis will be discussed below.

3.3.4. Reliability of the Paleomagnetic Results

Table 1 and Figure 2 show that the fold in the area of sections 2 and 3 is likely a typical syncline, while the fold in the area incorporating sections 4 and 5 is a plunging syncline with a plunging fold axis calculated to be in a direction Dp/Ip = 233°/26° (α_{95}p = 3.5°). Hence, it is necessary to apply two-step unfolding (Stewart & Jackson, 1995) to the sites from sections 4 and 5 to correct the fold plunge.

The combination of the Tak Fa and Khao Khwang formation data yields 11 sites passing McElhinny's (1964) fold test with ks/kg = 3.93 > $F(20, 20)_{95\%}$ = 2.12 and $F(20, 20)_{99\%}$ = 2.94, and yields a positive McFadden's (1990) fold test result (in situ ξ_2 = 9.878, tilt-corrected ξ_2 = 0.526 with ξ_2 critical value = 3.865 and 5.378 at 95% and 99% confidence level for N = 11). A stepwise tectonic correction (Watson & Enkin, 1993) yields an optimal clustering at 100.9 ± 5.8% unfolding level. Following application of the plunging fold correction to the sites from the Khao Khwang Formation, the ksp/kg of the combined Tak Fa and Khao Khwang formation results is raised from 3.93 to 6.18. The reversal test (McFadden & McElhinny, 1990) yields class C and class B results before and after application of the plunging fold correction. The data set is improved significantly

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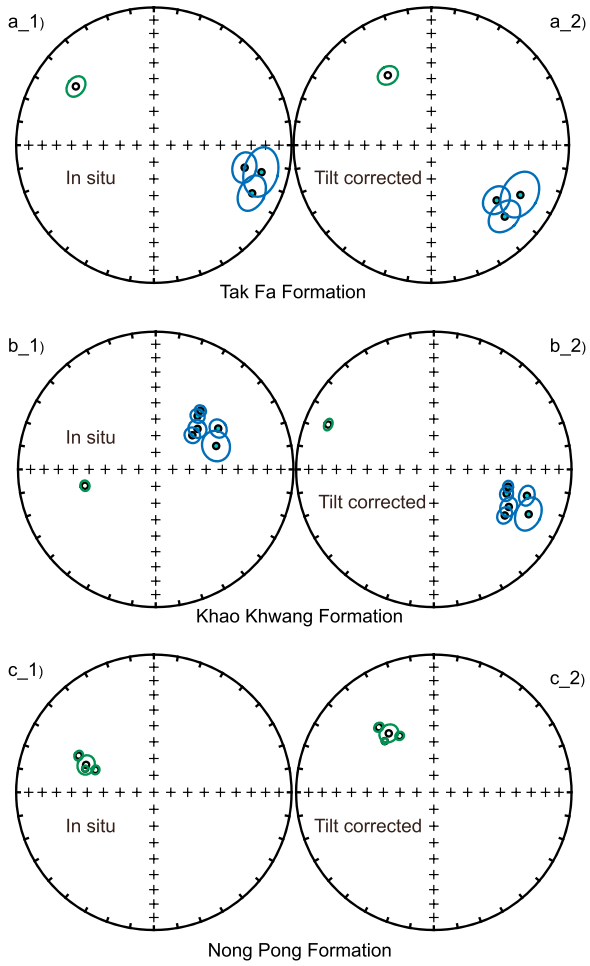


Figure 5. Equal-area projection of the directions of the ChRMs before (a_1, b_1, and c_1) and after (a_2, b_2, and c_2) tilt corrections for the Tak Fa, Khao Khwang and Nong Pong formations. Mean directions of normal (reversal) polarities are marked by solid (open) circles with blue (green) confidence ellipsoids.

following application of the plunging fold correction, and yields a mean direction (Fisher, 1953) of D_g/I_g = 79.7°/46.2° (k_g = 8.7, α_{95} = 16.4°, N = 11) before and D_s/I_s = 119.7°/34.2° (k_s = 34.2, α_{95} = 7.9°) after tilt correction, and D_{sp}/I_{sp} = 132.1°/33.8° (k_{sp} = 53.8, α_{95} = 6.3°) after the two-step unfolding correction. On the basis of these positive field tests it is very likely to be a prefolding magnetization (Figures 6a–6d).

Testing for the possibility that the ChRMs carried by the pyrrhotite-bearing sites are primary, we note that the data set including the pyrrhotite-bearing sites from the Nong Pong Formation and the magnetite-bearing sites from the Tak Fa and Khao Khwang formations collectively passes McElhinny's (1964) fold test with ks/kg = 2.80 and ksp/kg = 3.97 $>$ $F(28, 28)_{95\%}$ = 1.88 and $F(28, 28)_{99\%}$ = 2.46, and yields a positive McFadden's (1990) fold test (in situ ξ_2 = 12.252, tilt-corrected ξ_2 = 2.984, ξ_2 critical value = 4.510 and 6.305 at 95% and 99% confidence level for N = 15) with an optimal clustering at 88.4 ± 5.4% before and 89.7 ± 4.3% after the plunging fold correction (Watson & Enkin, 1993). However, the data set fails the reversal test (McFadden & McElhinny, 1990) with r = 13.5° $>$ r_critical = 11.0°. This could be due to regional relative rotation between these sections with the same cause possibly contributing to the deviation of the optimal clustering from 100% to 89.7% (Figures 6e–6h). Therefore, in short, the ChRMs carried by both magnetite-bearing and pyrrhotite-bearing samples are substantially consistent. It seems reasonable to include the ChRMs of all the 15 sites from the three formations in the calculation of mean result, since that the ages of the three formations are overlapped partially to each other, as a whole, the Early-Middle Permian.

Furthermore, we interpret that the southeast-downward direction paleomagnetic records (Figure 6) were acquired in reversal polarity epoch and the northwest-upward direction data acquired the ChRMs

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Table 1
Summary of sampling information and characteristic remanence from the Permian rocks in the Indochina Block

Site ID	Formation	λ_s (°N)	φ_s (°E)	Strike/dip	N	D_g (°)	I_g (°)	D_s (°)	I_s (°)	D_{sp} (°)	I_{sp} (°)	κ	α_{95} (°)	λ_p (°N)	φ_p (°E)
Th219	Tak Fa	15.3405	100.6014	156/31	10	307.0	−29.8	328.1	−40.0	—	—	72.8	5.7	40.7	320.6
Th226	Tak Fa	15.8148	100.9325	122/34	6	104.1	20.5	119.2	26.8	—	—	31.1	12.2	22.8	347.6
Th228	Tak Fa	15.8153	100.9321	132/39	6	116.1	21.6	134.2	26.5	—	—	55.2	9.1	35.7	340.0
Th229	Tak Fa	15.8155	100.9321	125/36	5	103.9	32.6	130.2	37.9	—	—	92.4	8.0	28.6	335.2
Th392	Khao Khwang	14.9012	101.2791	69/58	7	47.0	60.2	123.0	38.5	142.7	38.3	173.1	4.6	38.3	327.2
Th394	Khao Khwang	14.9012	101.2791	69/57	9	38.1	49.2	108.4	43.6	127.9	43.6	158.6	4.1	25.1	333.2
Th395	Khao Khwang	14.9012	101.2791	69/57	9	257.0	−46.2	293.0	−17.8	312.5	−17.8	315.9	2.9	37.1	347.2
Th397	Khao Khwang	14.9012	101.2791	69/57	5	68.8	51.1	115.3	24.8	134.8	24.8	80.1	8.6	37.3	341.6
Th398	Khao Khwang	14.9012	101.2791	69/57	10	56.8	44.9	105.8	30.4	125.2	30.4	87.2	5.2	27.5	343.4
Th401	Khao Khwang	14.9012	101.2791	69/57	11	45.9	55.2	116.6	39.0	136.1	39.0	75.2	5.3	33.3	331.5
Th402	Khao Khwang	14.9012	101.2791	69/57	10	37.3	45.4	103.1	43.9	122.6	43.9	228.7	3.2	21.0	335.7
Th403	Nong Pong	14.8255	101.3481	126/29	9	291.0	−52.5	328.8	−50.4	337.1	−50.4	424.8	2.5	39.0	306.7
Th404	Nong Pong	14.8255	101.3481	125/25	10	289.0	−46.2	316.2	−47.3	324.4	−47.3	807.5	1.7	34.7	319.8
Th405	Nong Pong	14.8255	101.3481	126/29	8	296.0	−39.1	319.8	−38.1	328.1	−38.1	421.6	2.7	42.1	322.9
Th406	Nong Pong	14.8255	101.3481	126/29	11	292.0	−45.9	322.6	−45.2	330.9	−45.2	72.4	5.4	39.7	315.7
	Mean of sites (Tak Fa + Khao Khwang)				11	79.7	46.2					8.7	16.4		
					11			119.7	34.2			34.2	7.9		
					11					132.1	33.8	53.8	6.3		
	Mean of sites (Tak Fa + Khao Khwang + Nong Pong)				15	88.9	47.1					10.2	12.6		
					15			124.9	37.6			28.6	7.3		
					15					136.3	37.2	40.5	6.1		
	Mean of VGPs (Tak Fa + Khao Khwang)				11							K = 69.1	A_{95} = 5.5	31.9	336.9
	Mean of VGPs (Tak Fa + Khao Khwang + Nong Pong)				15							K = 45.4	A_{95} = 5.7	34.1	331.7

Note. λ_s/φ_s = latitude and longitude of sampling site; N = number of samples or sites used in the mean calculation; Strike/dip = strike azimuth and dip of bed; D_g, I_g/D_s, I_s/D_{sp}, I_{sp} = declination and inclination before and after tilt correction and after plunging fold correction; κ and α_{95} = precision parameter and the radius of the cone of 95% confidence for mean direction; λ_p/φ_p = latitude and longitude of virtual geomagnetic pole (VGP); K and A_{95} = precision parameter and the radius of the cone of 95% confidence for formation mean VGP.
(1) McElhinny' fold test (1964):
Tak Fa + Khao Khwang: ksp/kg = 6.18 and ks/kg = 3.93 > $F(20, 20)_{95\%}$ = 2.12 and $F(20,20)_{99\%}$ = 2.94; indicate positive fold test at 95% and 99% confidence level.
Tak Fa + Khao Khwang + Nong Pong: ksp/kg = 3.97 and ks/kg = 2.80 > $F(28, 28)_{95\%}$ = 1.88 and $F(28, 28)_{99\%}$ = 2.46; indicate positive fold test at 95% and 99% confidence level.
(2) McFadden's fold test (1990):
Tak Fa + Khao Khwang: ξ_2 = 9.878 before and ξ_2 = 0.526 after tilt correction, critical ξ, at 95% = 3.865 and at 99% = 5.378, indicate positive fold test at 95% and 99% confidence level.
Tak Fa + Khao Khwang + Nong Pong: ξ_2 = 12.252 before and ξ_2 = 2.984 after tilt correction, critical ξ, at 95% = 4.510 and at 99% = 6.305, indicate positive fold test at 95% and 99% confidence level.
(3) Watson and Enkin's fold test (1993):
Tak Fa + Khao Khwang: optimal concentration at 100.9 ± 5.8% unfolding percentage indicate positive fold test at 95% confidence level.
Tak Fa + Khao Khwang + Nong Pong: optimal concentration at 89.7 ± 4.3% unfolding percentage indicate positive fold test at 95% confidence level.
(4) McFadden and McElhinny's reversal test (1990):
Tak Fa + Khao Khwang: pass reversal test at 95% confidence level in class C before and class B after plunging fold conrrection.
Tak Fa + Khao Khwang + Nong Pong: not pass reversal test at 95% confidence level.

occasionally in normal polarity epoch in the southern hemisphere. Because the Early-Middle Permian geomagnetic field was dominated by reversed geomagnetic polarity (Gradstein et al., 2012), which yielded south-downward dominated paleomagnetic directions in the southern hemisphere and north-downward dominated ones in the northern hemisphere. Otherwise, if we speculate the ChRMs were acquired in the northern hemisphere, the Indochina will be supposed to clockwise rotate for 136.3° (magnetic declination), which is not likely a reasonable tectonic movement scene. Moreover, the structure of the data set (southeast downward data are much more than the northwest upward ones) does not match with the reversed polarity dominated Early-Middle Permian geomagnetic field.

Strictly speaking, all tectonic events after the formation of the strata contributed to the folding now in different ways. Based on the prefolding origin of the Permian ChRMs in this study acquired prior to the regional tectonic events including the Indosinian, Middle Cretaceous, and Himalayan Orogeny, it is strongly suggested that this study has recovered a Permian paleo-field direction. The Permian strata are separated by the Indosinian unconformity from the Mesozoic sediments whose magnetic directions (e.g., Norian age of Late Triassic, Ds/Is = 43.0°/48.0°, α_{95} = 6.1°; Early Cretaceous, Ds/Is = 28.6°/39.6°, α_{95} = 3.1°) are distinctly

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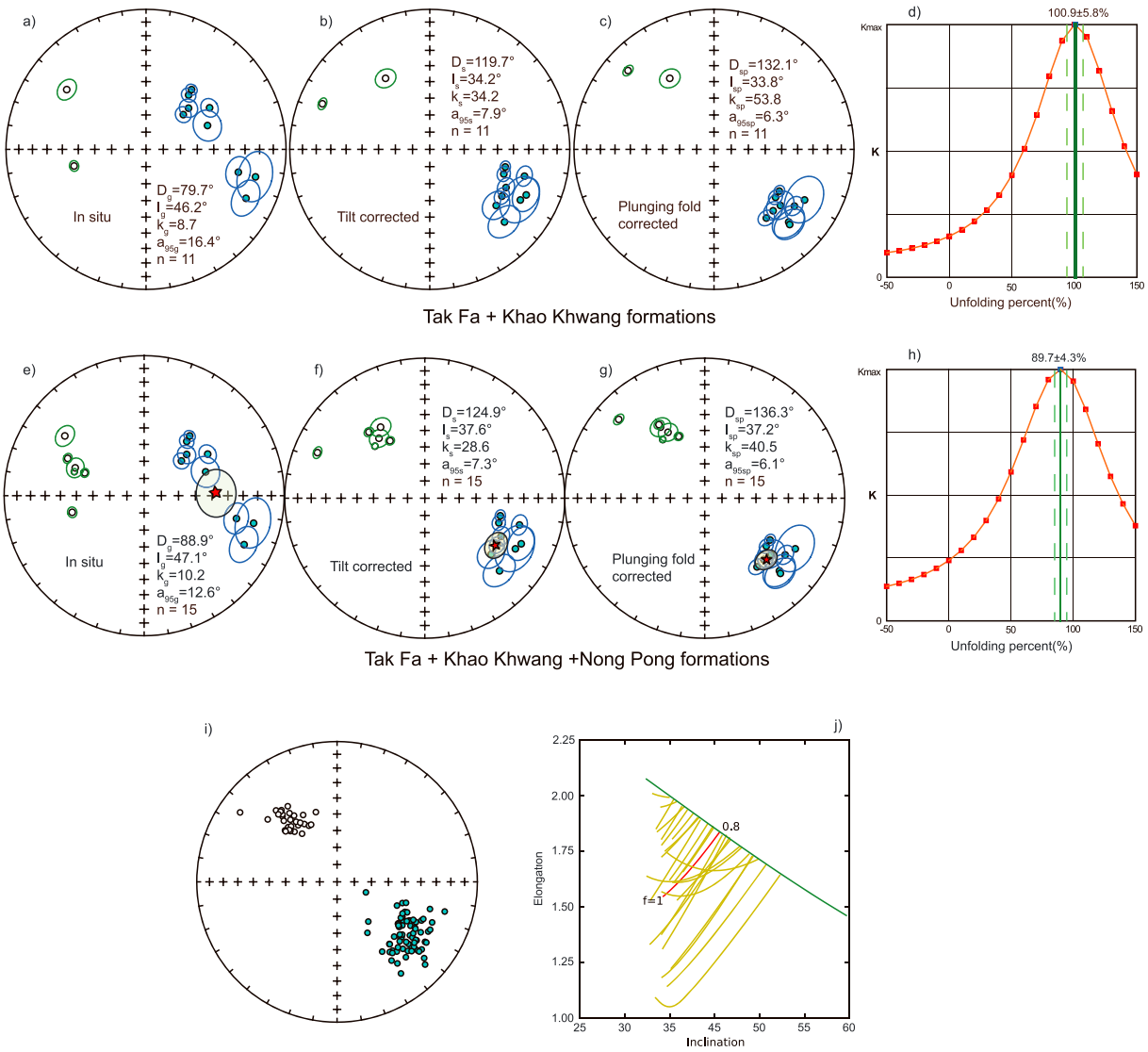


Figure 6. Equal-area projection of the site mean directions of the ChRMs before (a and e) and after tilt correction (b and f), and after plunging fold correction (c and g); (d and h) increment unfolding analysis using Watson and Enkin's (1993) method; (i) samples calculated with principle component analysis (PCA); (j) elongation-inclination modeling analysis. See text for discussion. $\mathbf{D_g}$, $\mathbf{I_g/D_s}$, $\mathbf{I_s/D_{sp}}$, $\mathbf{I_{sp}}$ = declination and inclination before and after tilt correction and after plunging fold correction; κ & α_{95}: precision parameter and the radius of the cone of 95% confidence for mean direction; n = number of sites used to calculate. Pmagpy software package are used to plot (Tauxe et al., 2016).

different (Yan et al., 2017), which rules out the possibility of remagnetization. The Stage I of the Indosinian Event corresponds to a considerable hiatus including much of the Upper Permian to Middle Triassic (Ridd et al., 2011).

4. Discussion

4.1. Is the Pyrrhotite Record Reliable?

The presence of pyrrhotite (Fe_7S_8) is increasingly being reported from sediments, although geochemical conditions required for the formation of pyrrhotite are poorly understood and many factors are considered to potentially contribute to the formation of ferromagnetic iron sulfides (greigite and pyrrhotite). These include the availability of fine iron-containing mineral and sulfate grains, bioactivity and specific redox conditions. It can be authigenic products from an early stage of diagenesis under anoxic sedimentary environments (Hall & Evans, 1995; Kao et al., 2004). It may also occur in a late stage of lithification as a record of syn-folding

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magnetization (Kars & Kodama, 2015; Weaver et al., 2002) while in some specific conditions it also may be detrital in origin (Horng et al., 1998; Horng & Roberts, 2006).

In this study, we tend to interpret the pyrrhotite as of detrital origin for two reasons. First, magnetite and pyrrhotite coexist in the limestone samples and record the same magnetization components as discussed in section 3.3.3, which strongly indicate that the ChRMs are reliable.

4.2. The Paleo-Positions of the Indochina Block

The Fisherian mean (Fisher, 1953) of the virtual geomagnetic poles of the Tak Fa and Khao Khwang formations yields a paleomagnetic pole at 31.9°N, 336.9°E (K = 69.1, A_{95} = 5.5°), which is not significantly different from that for all the three formations at 34.1°N, 331.7°E (K = 45.4, A_{95} = 5.7°; Table 1). Although the former pole resulted from a data set with more robust field test (section 4.2), we incline to use the later one as the paleomagnetic pole in the tectonic analyses discussed below because inclusion of the data from the Nong Pong Formation can potentially be more representative of the Early-Middle Permian age range.

The elongation-inclination method developed by Tauxe and Kent (2004) was used to estimate the likely inclination shallowing error on 105 samples whose directions were calculated with PCA (Kirschvink, 1980). A shallowing factor of f = 0.8 is produced after 1,000 bootstrap simulations (Figures 6i and 6j), which gives a corrected inclination of 45.6° with 95% confidence limits between 39.4° and 52.9°. The factor (f = 0.8) is used for the correction of the data set including 105 PCA-calculated and 12 GC-calculated samples. The pole was corrected from 34.1°N, 331.7°E (K = 45.4, A_{95} = 5.7°) to 31.2°N, 327.3°E (K = 45.4, A_{95} = 5.7°; Figure 6) and corresponds to a paleolatitude correction from 21.2 ± 5.7°S to 25.9 ± 5.7°S at the center of study area (15°N, 101°E), or 10.9 ± 5.7°S to 15.6 ± 5.7°S at the reference site (28°N, 100°E). The inclination shallowing effect is supposed to be not significant in this study on the consideration of the paleolatitude correction of ~4.7° relative to the uncertainty of 5.7°. We tend to be conservative in using the indistinctively E-I corrected result for the potential risk that the elongation is resulted from probable relative rotation between different sections.

Thus, the Early-Middle Indochina Block is interpreted to be located at a middle-low latitude (21.2 ± 5.7°S) position in the Southern Hemisphere (Figures 7 and 8a) consistent with the tropical restorations (Cocks & Torsvik, 2013; Scotese, 2004; Torsvik & Cocks, 2013; Ueno, 2003) but contrasting with the equatorial model (Collins, 2003; Ferrari et al., 2008; Metcalfe, 1996, 1999, 2001, 2002, 2011a, 2011b, 2013). This suggests that the Indochina Block had drifted away from the northern margin of Gondwana by the Early-Middle Permian (ca. 280 Ma; Figures 7 and 8a), followed by a ~5,000-km (~47° of latitudinal difference) northward transition until it merged to the southern margin of the Eurasian region in the Indosinian Orogeny Event by the Norian Stage of the Late Triassic (ca. 220 Ma; Figures 7 and 8b; Yan et al., 2017). This implies an average northward drift rate of ~9 cm/year relative to the Earth's rotation axis, and ~4.5 cm/year relative to Gondwana, because Gondwana had moved northward by ~23°during the same time interval (ca. 280 Ma to ca. 220 Ma; Torsvik et al., 2012). The estimation of ~4.5 cm/year is a cumulative result of the closing of the Paleo-Tethys and the opening of the Neo-Tethys (Li et al., 2016; Scotese, 2004).

4.3. The Paleogeographic Configuration and Drifting History of the Cathaysialand

Indochina was identified paleontologically to be of Cathaysia/Tethys affinity during the Carboniferous to Triassic or even Early Jurassic (Metcalfe, 2011a). The typical Cathaysian *Giagantopteris* floras were recognized in the Indochina Block as well as in the South China and North China blocks, and the endemic occurrence of *Pseudosweetognathus* on the Indochina and South China blocks was considered to be an evidence for their continental connection or close affinity (Metcalfe & Sone, 2008). The recognition of ophiolite, high-pressure metamorphic rocks (eclogite and blueschist) from central Qiangtang (Kapp et al., 2000, 2003; Zhai et al., 2007, 2009, 2011; Zhang et al., 2006) suggests the Longmucuo-Shuanghu Suture between the North Qiangtang and South Qiangtang blocks may be the westward extension of the Changning-Menglian Suture and represents the main suture zone of the Paleo-Tethys Ocean (Metcalfe, 2013; Zhai et al., 2011). Paleomagnetic data isolated from Late Carboniferous to Early Permian sedimentary rocks (Yang et al., 2017) and Early Permian (296.9 ± 1.9 Ma) volcanic rocks (Song et al., 2017) indicated similar paleolatitudinal position (30–34°S; Figure 7 and Table S2) for the North Qiangtang Block, which is in good agreement with the paleo-position of the South China Block as synthesized by Cocks and Torsvik (2013). Note that the synthesized apparent polar wander paths of the South China Block in Cocks and Torsvik (2013) should be used cautiously because inclination correction with a factor of 0.6 to all clastic sedimentary rocks and may lead to an unnecessary

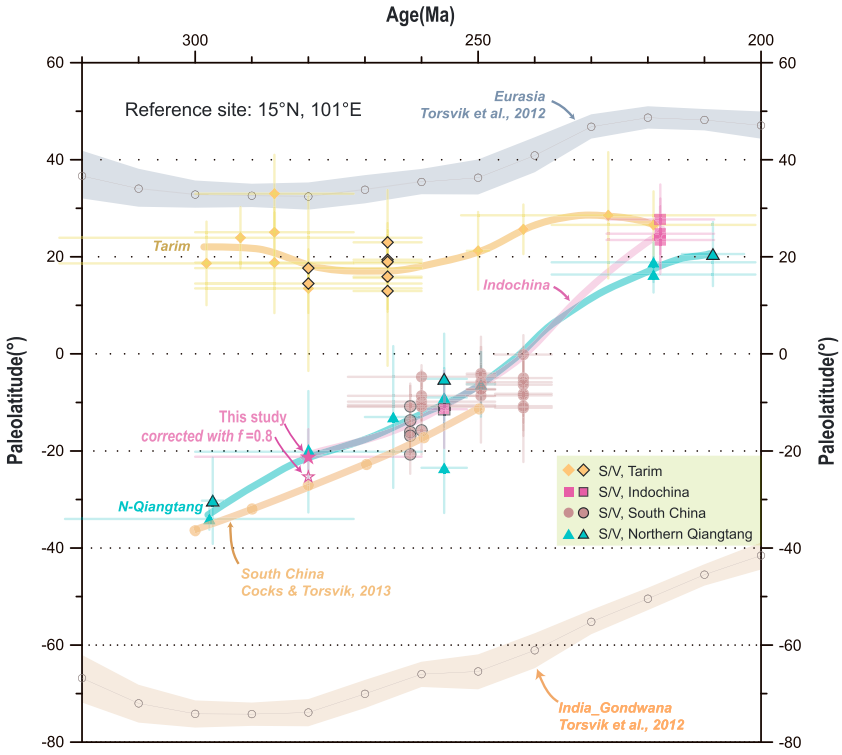


Figure 7. Paleolatitude evolution of Indochina, Northern Qiangtang, South China, and Tarim blocks. All data presented here are summarized in Table S2. (Indochina: Achache & Courtillot, 1985; Chi et al., 2016; Yan et al., 2017; Yang & Besse, 1993. Northern Qiangtang: Cheng et al., 2012, 2013; Huang et al., 1992; Song et al., 2012, 2015, 2017; Yang et al., 2017. North China: Embleton et al., 1996; Fang et al., 1988; Huang et al., 2001, 2008; Ma et al., 1992; Tan et al., 1991, 1995; Yang et al., 1991, 1998; Zhao et al., 1990. South China: Bai & Wu, 1997; Enkin, 1990; Enkin et al., 1992; Fang et al., 1990; Gilder et al., 1995; Heller et al., 1988, 1995; Huang et al., 1986; Huang & Opdyke, 1996, 1997; Li, Wang, et al., 1988; Li & Wang, 1989; Liu et al., 1987; Liu et al., 2012; Ma et al., 1993; Ma & Zhuang, 1989; Opdyke et al., 1986; Steiner et al., 1989; Su et al., 2005; Wang et al., 2013; Wu et al., 1999; Yang & Besse, 2001; Zhang et al., 2001; Zhao & Coe, 1989. Tarim: Bai et al., 1987; Fang, 1991; Fang et al., 1996, 1997, 2006; Gilder et al., 1996, 2008; Li, 1990; Li, McWilliams, et al., 1988; Li et al., 1989, 1995; McFadden et al., 1988; Sharps et al., 1989; Zhai et al., 1988. India of Gondwana and Eurasia: Torsvik et al., 2012.) Paleolatitudes of Gondwana_Indian and Eurasia are calculated from Torsvik et al., 2012, using the online paleomagnetic toolkit on Paleomagnetism.org. S/V: paleomagnetic data revealed from sedimentary or volcanic rocks.

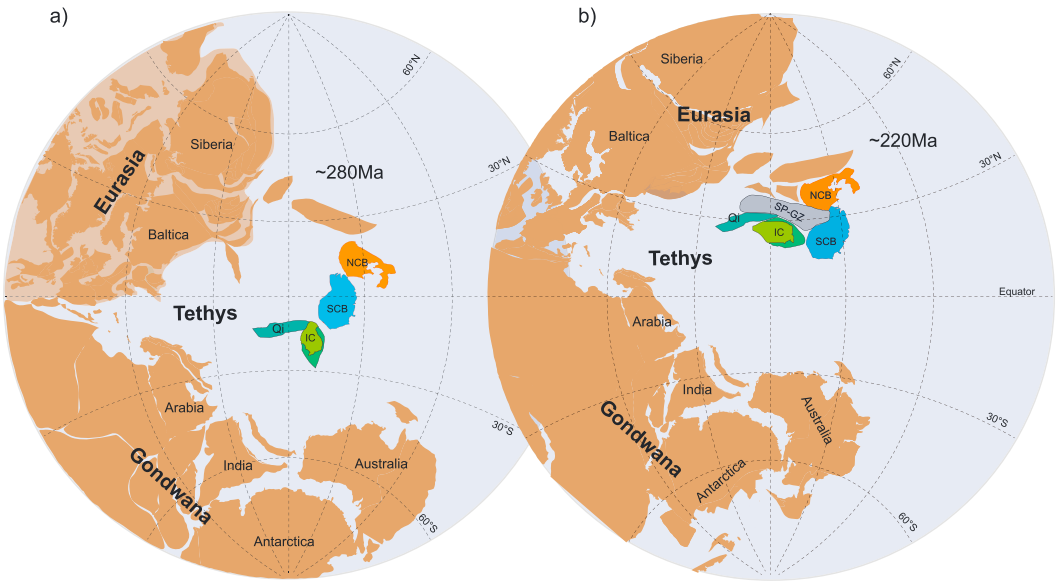


Figure 8. Reconstruction of the Indochina (IC), North Qiangtang (QI), South China (SCB), and North China (NCB) blocks in the Pangea supercontinent at Early-Middle Permian (a) and Late Triassic (b). Data used are include in Table S2.

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offset for some data sets. During the Early to Middle Permian, the new paleomagnetic data obtained from the Indochina Block in this study suggests a ~21°S paleolatitude, which is almost the same as that of the North Qiangtang Block (~20°S; Cheng et al., 2012). The paleolatitude can be corrected to ~26°S with a flatting factor of 0.8 as discussed in section 4.3 and matches well with the South China result. A preliminary Late Permian result from 33 basaltic samples in 4 sites in Northwest Vietnam (Chi et al., 2016) yielded a paleolatitude of ~12°S for the Indochina Block, which is consistent with the plentiful Late Permian to Early Triassic results of the South China and North Qiangtang blocks. In summary, paleobiogeographic affinities and concurrent drifting history of the Indochina, North Qiangtang, and South China blocks implies they probably belonged to an integral Permian mainland in the Paleo-Tethys, the Cathaysialand.

Hence, the new Early-Middle Permian paleomagnetic data obtained in this study provides a robust constraint on the Permian paleo-position of the Indochina Block and the configuration and northward movement history of the Cathaysialand during Permian before they amalgamated to the Eurasian mainland by the Late Triassic (Figures 7 and 8).

5. Conclusions

We provide a new high-quality Early-Middle paleomagnetic pole from central Thailand for the Indochina Block, which indicates a paleolatitude of 21.2 ± 5.7°S for the center of study area (15°N, 101°E). The tropical pattern rather than the equatorial pattern of late Paleozoic restoration of Indochina is supported. We confirm that the Indochina Block had drifted away from the northern margin of Gondwana by the Early-Middle Permian. The average rate of the Indochina Block drifting away from Gondwana is estimated as ~4.5 cm/year and a concurrent drifting history is supposed for the Indochina, North Qiangtang, and South China blocks. The model of Cathaysialand including the South China, Indochina, and North Qiangtang blocks is quantitatively supported.

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